

LAB 3 JAVA

1. Create a class Login with private username and password. Password should be at least 8 characters (validate in setter)

```
login > J LoginTest.java 1 X
login > J LoginTest.java > Java > Login > getPassword()
1  class Login {
2      private String username;
3      private String password;
4
5      public void setUsername(String username) {
6          this.username = username;
7      }
8
9      public void setPassword(String password) {
10         if (password.length() >= 8) {
11             this.password = password;
12         } else {
13             System.out.println("Password must be at least 8 characters");
14         }
15     }
16
17     public String getUsername() {
18         return username;
19     }
20
21     public String getPassword() {
22         return password;
23     }
24 }
25
26 public class LoginTest {
27     Run main | Debug main | Run | Debug
28     public static void main(String[] args) {
29         Login l = new Login();
30         l.setUsername(username: "admin");
31         l.setPassword(password: "admin123");
32
33         System.out.println("Username: " + l.getUsername());
34         System.out.println("Password: " + l.getPassword());
35     }
}
```

Output:

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Password: admin123
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\login> cd "c:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\login" ; if ($?) { javac LoginTest.java }
Username: admin
Password: admin123
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\login>
```

2. Create a class Employee with fields id, salary, and designation Use encapsulation to set and get the values

```
login > EmployeeTest.java > EmployeeTest.java > Language Support for Java(TM) by Red Hat > EmployeeTest
1  class Employee {
2      private int id;
3      private double salary;
4      private String designation;
5
6      public void setId(int id) {
7          this.id = id;
8      }
9
10     public void setSalary(double salary) {
11         this.salary = salary;
12     }
13
14     public void setDesignation(String designation) {
15         this.designation = designation;
16     }
17
18     public int getId() {
19         return id;
20     }
21
22     public double getSalary() {
23         return salary;
24     }
25
26     public String getDesignation() {
27         return designation;
28     }
29 }
30
31 public class EmployeeTest {
32     Run main | Debug main | Run | Debug
33     public static void main(String[] args) {
34         Employee e = new Employee();
35         e.setId(id: 101);
36         e.setSalary(salary: 45000);
37         e.setDesignation(designation: "Software Developer");
38
39         System.out.println("Employee ID: " + e.getId());
40         System.out.println("Salary: " + e.getSalary());
41         System.out.println("Designation: " + e.getDesignation());
42     }
43 }
```

Output:

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise> cd "c:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\login\EmployeeTest."
; if ($?) { javac EmployeeTest.java } ; if ($?) { java EmployeeTest }
Employee ID: 101
Salary: 45000.0
Designation: Software Developer
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\login\EmployeeTest.java>
```